

LJUNGBYHED, Sweden

WMO: 020001

Lat: 56.083N

Lon: 13.213E

Elev: 140

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.6	14.5	4.7	7.1	10.9	10.4	9.4	17.2	30.2	40.8	27.1	39.4	2.1	110	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	82.3	66.3	78.7	65.5	75.2	63.7	69.1	77.4	67.2	74.9	65.4	72.6	9.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.5	72.7	64.5	91.2	70.4	62.6	85.5	68.4	33.2	77.5	31.6	74.9	30.3	72.5	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.9	21.5	18.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB	-0.1	72.2	8.8	2.3	-6.4	73.8	-11.6	75.2	-16.5	76.5	-22.9	78.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	32.4	31.9	36.0	44.6	52.0	57.7	63.4	61.9	55.0	48.2	40.8	34.4
	DBStd	12.78	8.14	7.02	6.96	6.17	6.19	5.26	5.01	4.60	4.90	6.21	6.61	8.83
	HDD50	2605	546	507	435	184	48	4	0	0	14	107	277	483
	HDD65	6788	1011	927	900	612	406	225	92	119	301	522	725	948
	CDD50	1371	0	0	1	22	108	236	417	369	164	51	3	0
	CDD65	78	0	0	0	0	2	8	44	23	1	0	0	0
	CDH74	976	0	0	0	4	47	149	565	201	10	0	0	0
	CDH80	197	0	0	0	0	2	24	142	29	0	0	0	0
	WSAvg	8.5	9.7	9.4	9.5	8.7	8.3	8.2	7.2	7.2	8.0	8.3	8.6	8.7
	PrecAvg	33.3	2.8	2.3	1.9	1.6	2.2	3.3	3.3	3.7	3.1	3.2	2.9	3.0
(15) Precipitation	PrecMax	44.1	6.8	5.9	4.6	3.0	4.6	7.4	9.0	7.8	6.9	7.0	5.9	6.0
	PrecMin	23.1	0.2	0.4	0.6	0.3	0.9	0.0	0.1	0.9	0.5	1.4	0.9	0.7
	PrecStd	4.8	1.6	1.1	1.0	0.9	1.0	1.8	2.1	1.9	1.6	1.4	1.1	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	48.0	58.7	71.5	79.0	83.5	87.8	82.9	75.4	66.2	55.3	51.4
		MCWB	48.1	44.0	48.3	56.9	64.3	66.1	68.3	66.4	64.9	59.8	53.4	49.8
	2%	DB	46.5	45.1	52.0	66.2	74.8	78.5	84.1	79.1	69.9	60.9	53.2	48.5
		MCWB	45.6	43.1	45.8	53.8	60.8	64.5	67.6	65.9	62.4	57.6	51.4	47.4
	5%	DB	44.6	43.3	48.3	61.4	70.0	73.6	80.5	75.4	66.4	58.9	51.4	46.5
		MCWB	43.4	41.7	44.2	50.8	58.2	62.1	66.3	64.8	60.1	56.2	49.5	45.7
	10%	DB	42.7	41.3	46.0	57.2	66.1	69.7	76.7	71.7	64.0	57.0	49.6	44.8
		MCWB	41.4	39.8	43.1	48.9	56.0	60.6	64.8	63.5	58.7	54.6	48.3	44.0
	0.4%	WB	48.3	45.4	50.9	59.0	66.9	69.7	72.7	70.7	65.8	61.1	54.2	50.1
		MCDB	49.3	46.3	55.0	68.7	76.5	78.7	83.0	77.7	72.5	64.9	55.1	50.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.4	43.8	47.0	54.9	62.9	65.8	69.8	67.7	63.3	58.4	51.4	47.4
		MCDB	45.9	44.8	50.2	64.1	71.0	74.4	79.0	74.3	68.7	60.7	52.8	48.1
	5%	WB	43.1	41.8	44.9	52.1	59.9	63.7	68.2	65.9	61.1	56.6	49.7	45.5
		MCDB	43.9	43.0	47.2	58.8	67.3	71.6	76.5	72.4	65.3	58.6	50.7	46.1
	10%	WB	41.2	39.7	42.7	49.8	57.1	61.4	66.4	64.3	59.3	55.0	48.0	43.8
		MCDB	42.1	41.2	45.1	55.8	64.2	68.1	73.9	70.3	62.7	56.8	48.8	44.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	8.8	13.1	18.6	20.0	18.0	20.0	17.0	15.0	12.4	8.9	8.3
		MCDBR	7.0	9.7	17.4	27.2	28.8	26.8	28.3	23.8	19.9	14.0	8.5	7.7
		MCWBR	7.1	8.2	12.4	16.3	16.1	14.7	13.7	11.9	12.9	10.7	7.5	7.7
	5% WB	MCDBR	7.2	8.8	14.9	22.8	23.3	23.2	22.5	19.7	18.2	13.1	7.8	7.7
		MCWBR	7.4	7.9	11.6	14.7	14.4	14.1	12.7	11.7	12.6	10.6	7.4	7.9
(40) Clear-Sky Solar Irradiance	taub		0.308	0.332	0.354	0.390	0.387	0.380	0.403	0.403	0.371	0.355	0.334	0.301
	taud		2.256	2.280	2.301	2.270	2.330	2.374	2.332	2.337	2.412	2.386	2.318	2.242
	Ebn at Noon		186	229	255	262	270	274	264	255	248	219	175	157
	Edh at Noon		17	25	31	37	37	36	36	34	28	22	16	14
(44) All-Sky Solar Radiation	RadAvg		136	324	757	1286	1669	1763	1636	1316	928	450	171	92
	RadStd		26	52	80	166	163	145	197	106	100	67	31	10

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-295	N/A	N/A

Nomenclature: See separate page